1. Book and Author (Containment with Array of Objects + Utility Class)

Design a C++ program to manage a **Library System** using **containment** and a separate utility class for operations.

Requirements

1 Create class Author

Data Members:

- authorName
- country

Member Functions:

- acceptAuthor()
- displayAuthor()

---

2 Create class Book

Data Members:

- bookId
- bookTitle
- price
- Contains an object of class Author (Containment)

Member Functions:

- acceptBook()
- displayBook()

This demonstrates a **"has-a" relationship** (Book has an Author).

---

3 Create class PrintToScreen(Utility class)

This class should handle all array operations.

Static Member Functions:

1. acceptBooks(Book b[])
    - Accept details for an array of Book objects.
2. displayBooks(Book b[])
    - Display all book details including author information.
3. searchById(Book b[], int id)
    - Search and display book details by bookId.
    - Display message if not found.
4. searchByAuthor(Book b[], string authorName)
    - Search and display books written by a specific author.

4️⃣ In main() Function

- Create an array of 3 Book objects.
- Call methods from PrintToScreen:
    - acceptBooks()
    - displayBooks()
    - searchById()
    - searchByAuthor()

---

2. Problem Statement: Healthcare Management System

(Hierarchical Inheritance in C++)

Design a C++ program to demonstrate **Hierarchical Inheritance** using a Healthcare Management System.

🔷 Scenario

A hospital manages different types of medical staff such as:

- Doctor
- Nurse
- LabTechnician

All medical staff share common information, but each role has its own specific responsibilities and details.

Requirements

1 Create a Base Class MedicalStaff

Data Members:

- staffId
- name
- age
- contactNumber

Member Functions:

- acceptStaffDetails()
- displayStaffDetails()

---

2 Create Derived Class Doctor (inherits from MedicalStaff)

Additional Data Members:

- specialization
- consultationFee
- experience (in years)

Member Functions:

- acceptDoctorDetails()
- displayDoctorDetails()
- Calculate yearly income (consultationFee × estimated patients per month × 12)

---

3 Create Derived Class Nurse (inherits from MedicalStaff)

Additional Data Members:

- shift (Day/Night)
- wardAssigned
- salary

Member Functions:

- acceptNurseDetails()
- displayNurseDetails()

4️⃣ Create Derived Class LabTechnician (inherits from MedicalStaff)

Additional Data Members:

- labType
- numberOfTestsHandledPerDay
- salary

Member Functions:

- acceptLabTechnicianDetails()
- displayLabTechnicianDetails()

🔷 In main() Function

- Create one object each of:
    - Doctor
    - Nurse
    - LabTechnician
- Accept and display their complete details.